

Second Best Minimum Spanning Tree

A Minimum Spanning Tree T is a tree for the given graph G which spans over all vertices of the given graph and has the minimum weight sum of all the edges, from all the possible spanning trees. A second best MST T' is a spanning tree, that has the second minimum weight sum of all the edges, from all the possible spanning trees of the graph G .

Observation

Let T be the Minimum Spanning Tree of a graph G . It can be observed, that the second best Minimum Spanning Tree differs from T by only one edge replacement. (For a proof of this statement refer to problem 23-1 [here](#)).

So we need to find an edge e_{new} which is not in T , and replace it with an edge in T (let it be e_{old}) such that the new graph $T' = (T \cup \{e_{\text{new}}\}) \setminus \{e_{\text{old}}\}$ is a spanning tree and the weight difference $(e_{\text{new}} - e_{\text{old}})$ is minimum.

Using Kruskal's Algorithm

We can use Kruskal's algorithm to find the MST first, and then just try to remove a single edge from it and replace it with another.

1. Sort the edges in $O(E \log E)$, then find a MST using Kruskal in $O(E)$.
2. For each edge in the MST (we will have $V-1$ edges in it) temporarily exclude it from the edge list so that it cannot be chosen.
3. Then, again try to find a MST in $O(E)$ using the remaining edges.
4. Do this for all the edges in MST, and take the best of all.

Note: we don't need to sort the edges again in for Step 3.

So, the overall time complexity will be $O(E \log V + E + V E) = O(V E)$.